

Name: _____ Partner: _____ Period: _____

Function Makeover Studio

A design studio applies transformation rules to “restyle” functions. The base function f is defined by the table below. Your team must construct transformed versions, match transformations to descriptions, and reverse-engineer transformations from data.

Base function f :

x	-2	-1	0	1	2
$f(x)$	4	1	0	1	4

Tier R — Remediate

Three students each applied a different transformation to f . Their tables are shown below. Match each table to the correct description by writing the letter in the blank.

Descriptions: **A.** $g(x) = f(x) - 3$ **B.** $g(x) = f(x + 2)$ **C.** $g(x) = 2f(x)$

x	-2	-1	0	1	2
Table 1: $g(x)$	1	-2	-3	-2	1

Matches description: _____

x	-2	-1	0	1	2
Table 2: $g(x)$	8	2	0	2	8

Matches description: _____

x	-2	-1	0	1	2
Table 3: $g(x)$	0	1	4	9	16

Matches description: _____

Tier A — Apply

Using the base function f from the scenario, construct the table of values for $g(x) = 2f(x - 1) + 3$. Show how you compute each output.

x	$x - 1$	$f(x - 1)$	$2f(x - 1)$	$g(x) = 2f(x - 1) + 3$
-1	_____	_____	_____	_____
0	_____	_____	_____	_____
1	_____	_____	_____	_____
2	_____	_____	_____	_____
3	_____	_____	_____	_____

State the domain and range of g based on your table.

Domain of g : _____ Range of g : _____

Tier E — Extend

A second function h is defined by the table below. By comparing h to the base function f , identify all transformations applied to f to produce h , and write $h(x)$ in terms of f .

x	-2	-1	0	1	2
$f(x)$	4	1	0	1	4
$h(x)$	-7	-1	1	-1	-7

1. Compare outputs: how does $h(x)$ relate to $f(x)$ at each x -value?

2. Write $h(x)$ in terms of $f(x)$. Identify a and k in $g(x) = af(x) + k$.

$h(x) =$ _____, so $a =$ _____ and $k =$ _____

3. Describe in words all transformations that map f onto h .
